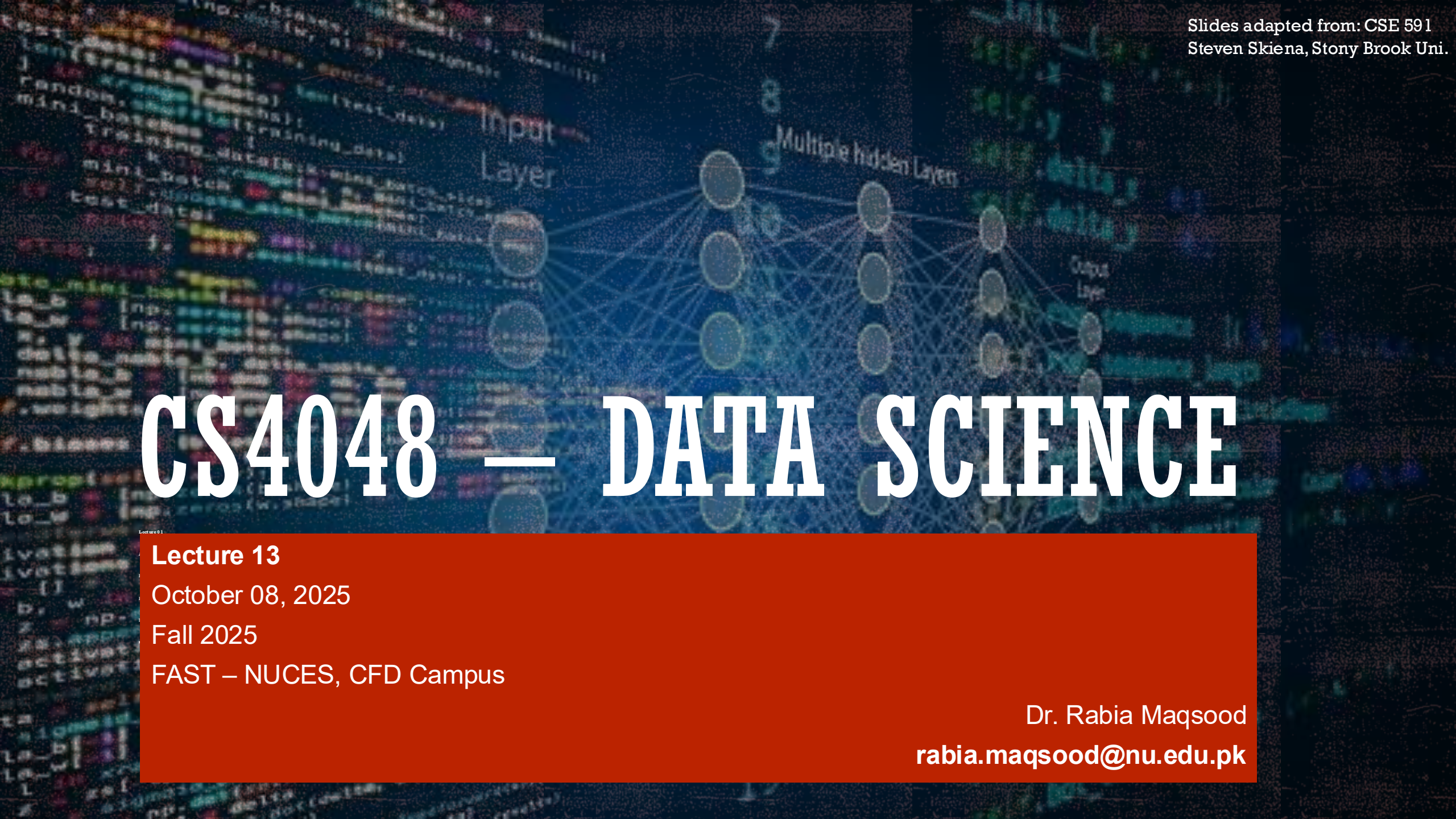


CS4048 — DATA SCIENCE



Lecture 13

October 08, 2025

Fall 2025

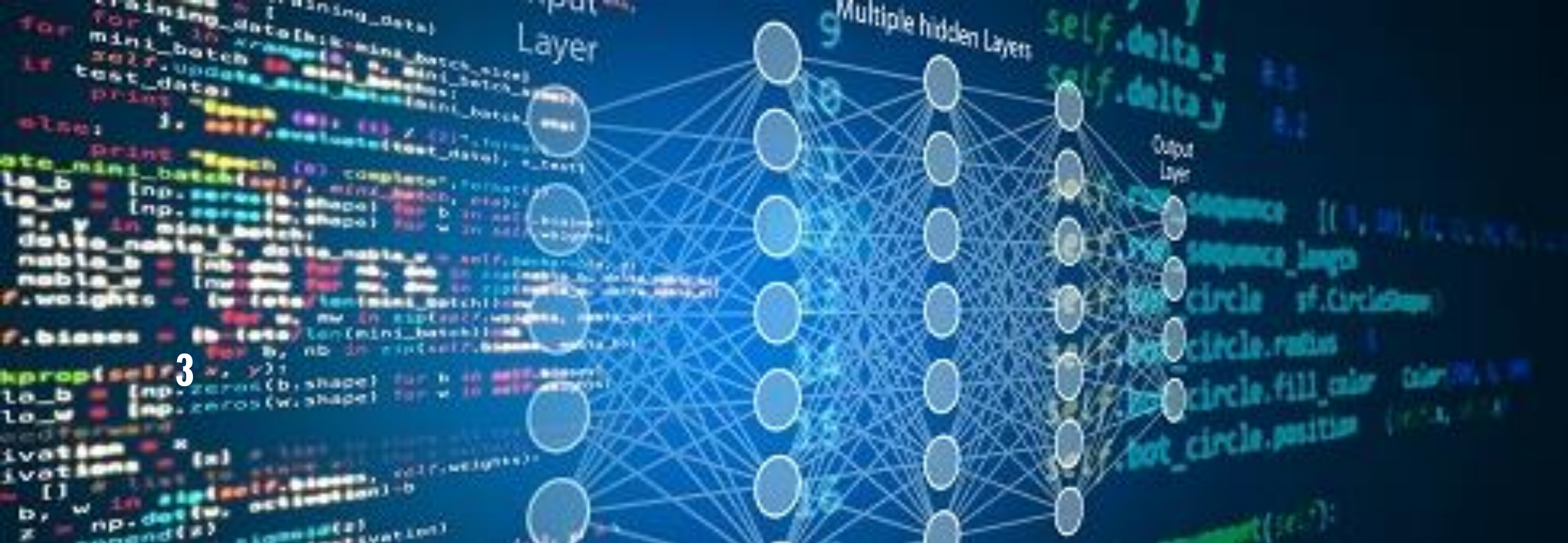
FAST – NUCES, CFD Campus

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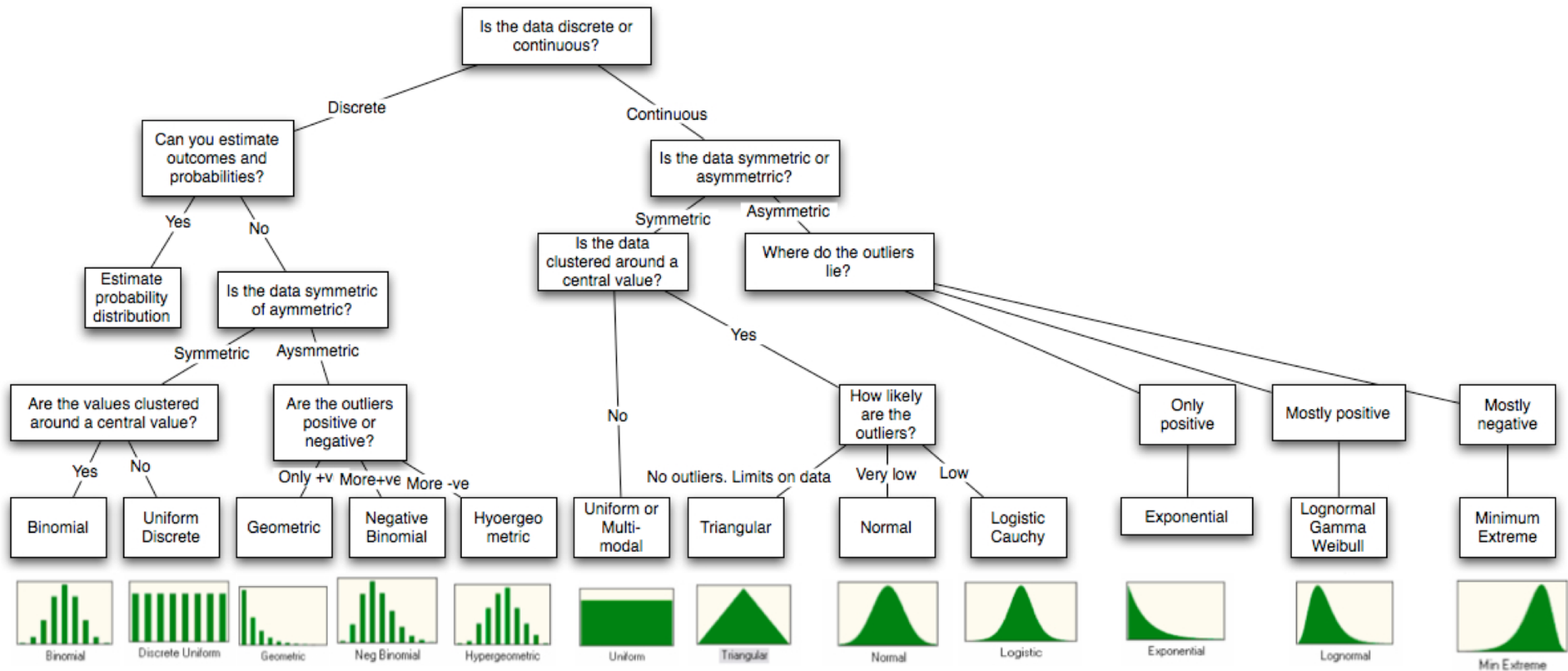
TODAY'S TOPICS

- Inferential statistics: introduction
- Probability distributions: overview
 - Binomial, Poisson Normal, Log normal,
 - Center limit theorem
- Data sampling methods



DATA DISTRIBUTIONS

Figure 6A.15: Distributional Choices

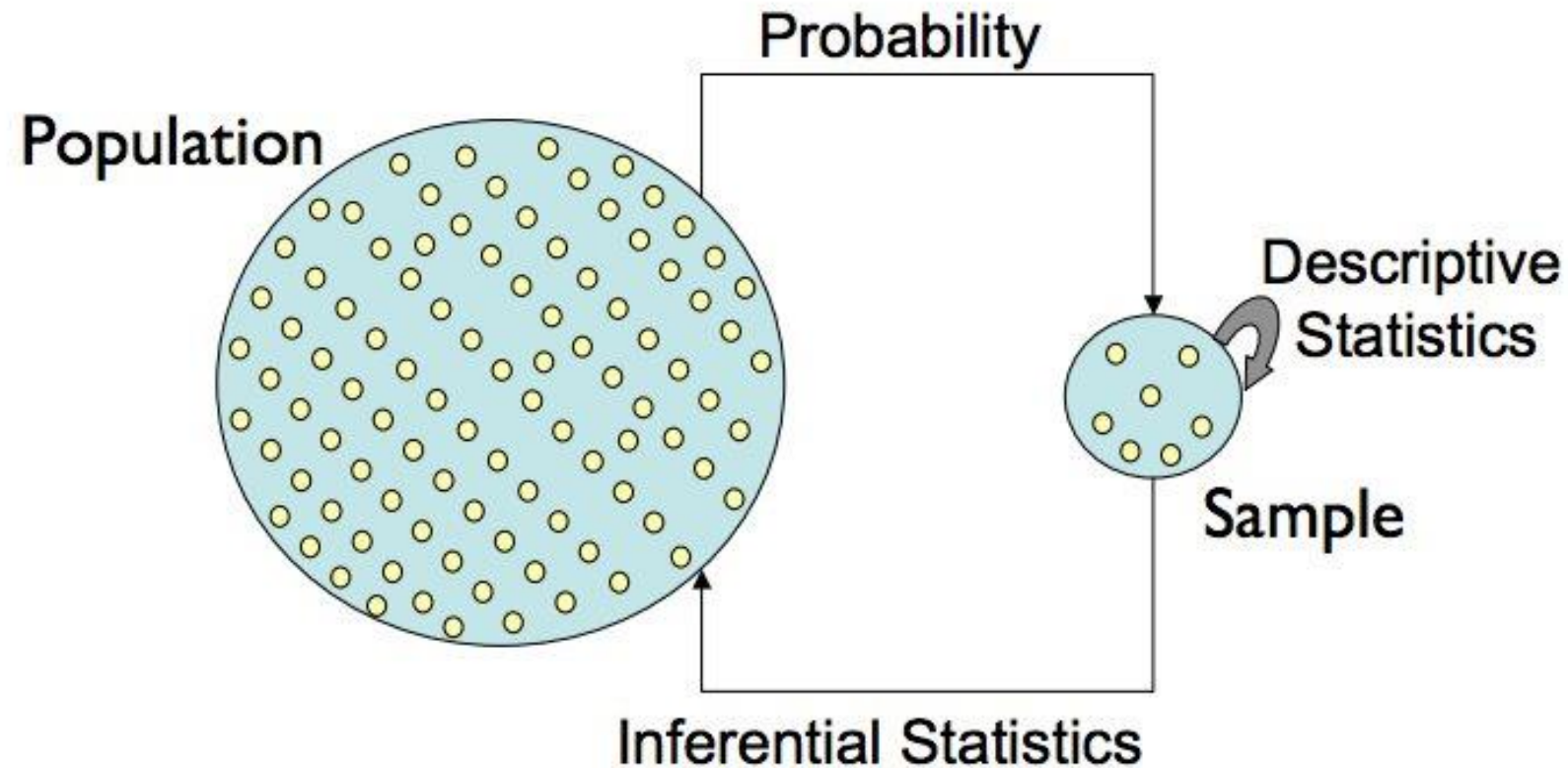


STATISTICS AND DATA SCIENCE

“A data scientist is someone who knows more statistics than a computer scientist and more computer science than a statistician.”

- Josh Blumenstock (Univ. of Washington)

THE CENTRAL DOGMA OF STATISTICS



PROBABILITY

- Probability is a numerical measure that assesses the likelihood of occurrence of an event.
- Probability applications are ubiquitous in data science since many decisions in business, science, and engineering are based on probability considerations.

BASIC TERMINOLOGIES

- An **outcome** is the result of a single trial in a probability experiment.
- The **sample space** is the set of all possible outcomes in a probability experiment.
- An **event** is some subset of the sample space.
 - For example, an event could be rolling an even number on a six-sided die. This event corresponds to three outcomes, namely rolling a 2, 4, or 6 on the die.

PROBABILITY CALCULATIONS

- Two approaches:
 1. Relative frequency probability
 2. Theoretical probability

RELATIVE FREQUENCY PROBABILITY

- Relative frequency probability is a method of **determining the likelihood of an event occurring** based on the **observed frequency of its occurrence in a given sample or population**.
- The probability of Event A , denoted as $P(A)$, is calculated based on data that has been collected from the experiment

$$P(A) = \frac{\text{number of times Event } A \text{ has occurred}}{\text{number of times the procedure was repeated}}$$

- Example: using the data collected from the past two years, there were 200 surgeries performed, with 188 successes. So, the probability of a successful knee surgery can then be calculated as: $188/200 = 0.94$

THEORETICAL PROBABILITY

- Theoretical probability is the method used when the outcomes in a probability experiment are equally likely — that is, under theoretical conditions.
- Theoretical probability considers all the possible outcomes for an experiment that are known ahead of time so that past data is not needed in the calculation for theoretical probability.

$$\text{Theoretical Probability} = \frac{\text{number of outcomes for the event of interest}}{\text{total number of outcomes in the sample space}}$$

- Example: the theoretical probability of rolling an even number when rolling a six-sided die is (which is , or 0.5). There are 3 outcomes corresponding to rolling an even number, and there are 6 outcomes total in the sample space.

PROBABILITY DISTRIBUTIONS

- Probability distributions are used to model various scenarios to help with probability analysis and predictions, and they are used extensively to help formulate probability-based decisions.

PROBABILITY DISTRIBUTION: TERMINOLOGIES

- **Random variable:** a variable where a single numerical value is assigned to a specific outcome from an experiment.
 - Variable x denoting a random draw from a pile of cards
- **Discrete random variable:** a random variable is considered discrete if there is a finite or countable number of values that the random variable can take on.
 - Typically, a discrete random variable is the result of a count of some kind.
 - No. of cars in a parking lot should be represented by a whole number/discrete value since 15.7 cars parked doesn't make sense.
- **Continuous random variable:** a random variable is considered continuous if the value of the random variable can take on any value within an interval.
 - Typically, a continuous random variable is the result of a measurement of some kind.
 - Avg. parking time of a car is 6.58 hrs.

PROBABILITY DISTRIBUTIONS

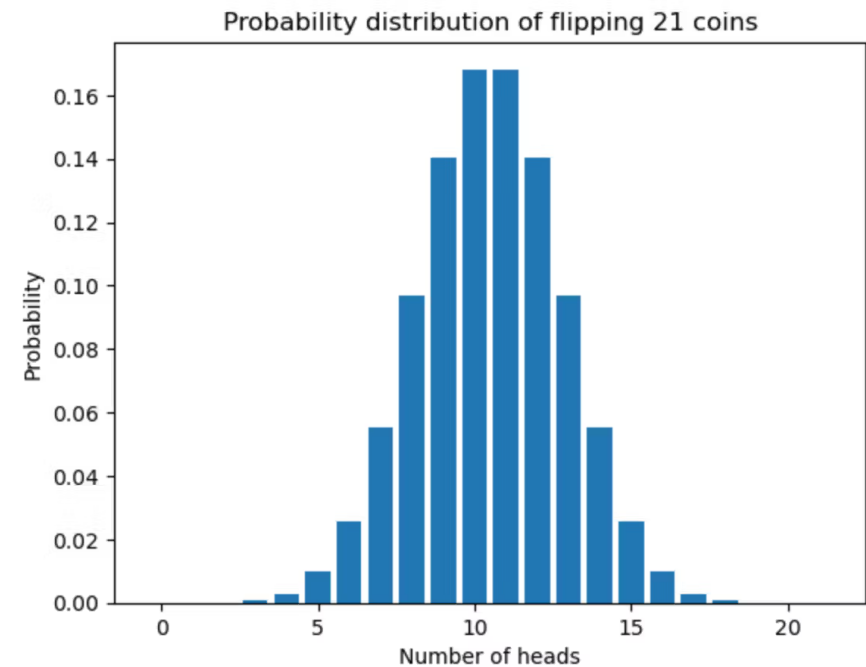
- Every observed random variable has a particular frequency/probability distribution.
- Nature of a variable also determines its probability distribution.
- Discrete probability distributions are associated with variables that take on a finite or countably infinite number of distinct values.
- Continuous probability distributions deal with random variables that can take on any value within a given range or interval.
- Also, different statistical methods are used to analyze each type.

DISCRETE PROBABILITY DISTRIBUTIONS

- Discrete random variables are of interest in many data science applications, and there are several probability distributions that apply to discrete random variables.
- Two primary distributions include:
 - Binomial distribution
 - Poisson distribution

BINOMIAL DISTRIBUTION

- The binomial distribution is used in applications where there are two possible outcomes for each trial in an experiment and the two possible outcomes can be considered as success or failure.
 - Real-life examples? (other than a coin flip)



BINOMIAL DISTRIBUTION

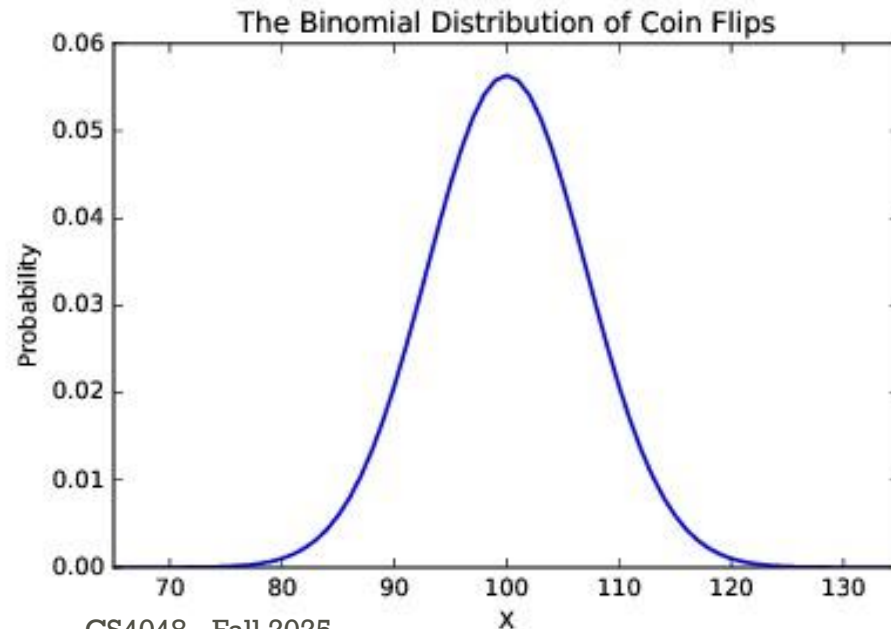
Experiments consist of n *identical, independent* trials which have two possible outcomes, with probabilities p and $(1-p)$ like heads or tails.

$$P\{X = x\} = \binom{n}{x} p^x (1-p)^{n-x}$$

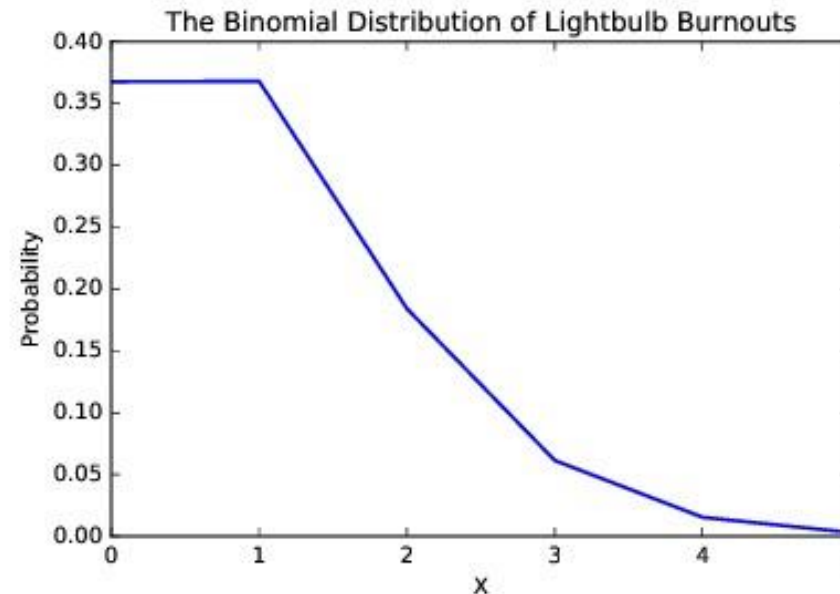
PROPERTIES OF BINOMIAL DISTRIBUTIONS

Discrete, but bell (or half-bell) shaped

Coin flips: $p=0.5$ $n=200$



Lightbulb burnouts:
 $p=0.001$ $n=1000$



The distribution is a function of n and p .

POISSON DISTRIBUTION

- Besides a binomial distribution, a data scientist might be interested to know the probability of a certain number of occurrences for a random variable in a specific interval, such as an interval of time.
- For example, a website developer might be interested in knowing the probability that a certain number of users visit a website per minute.
- The **Poisson distribution** is applied when counting the number of occurrences in a certain interval.
 - The random variable then counts the number of occurrences in the interval where an interval could represent time, area, volume, etc.

POISSON DISTRIBUTION

- To calculate the probability of x successes, the Poisson probability formula can be used as:

$$P(x) = \frac{\mu^x e^{-\mu}}{x!}, \text{ where } x = 0, 1, 2, \dots$$

Where:

μ is the average or mean number of occurrences per interval

e is the constant 2.71828...

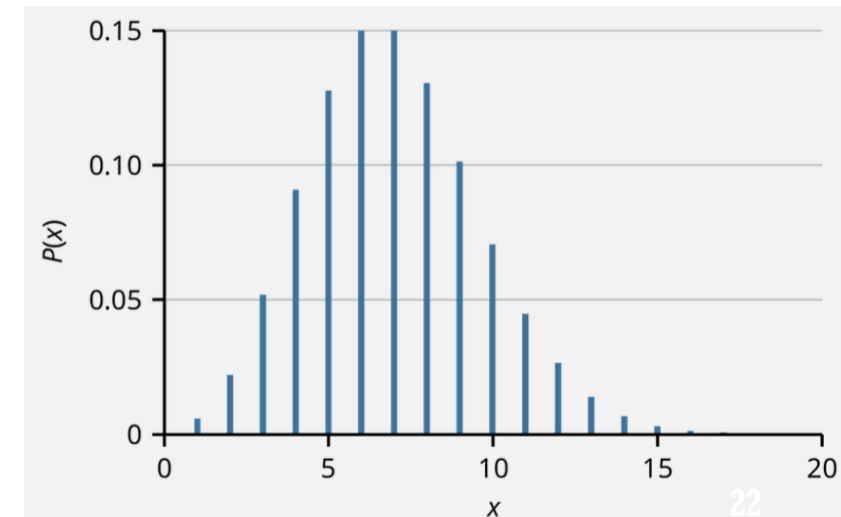


Figure 3.6 Poisson Distribution for $\mu = 7$

CONTINUOUS PROBABILITY DISTRIBUTION

- Many probability distributions apply to continuous random variables.
- These distributions rely on determining the probability that the random variable falls within a distinct range of values, which can be calculated using a **probability density function (PDF)**.
- For example, to determine the probability that a salary falls between \$50,000 and \$70,000, we can calculate the area under the probability density function between these two salaries.

NORMAL DISTRIBUTION

- The normal distribution is a continuous probability distribution that is symmetrical and bell-shaped.
 - The bell-shaped distribution of height, IQ, etc.
- It is used when the frequency of data values decreases with data values above and below the mean.
- Completely parameterized by mean and standard deviation:

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma}} e^{-(x-\mu)^2 / 2\sigma^2}$$

- Not all bell-shaped distributions are normal but it is generally a reasonable start.

PROPERTIES OF THE NORMAL DISTRIBUTION

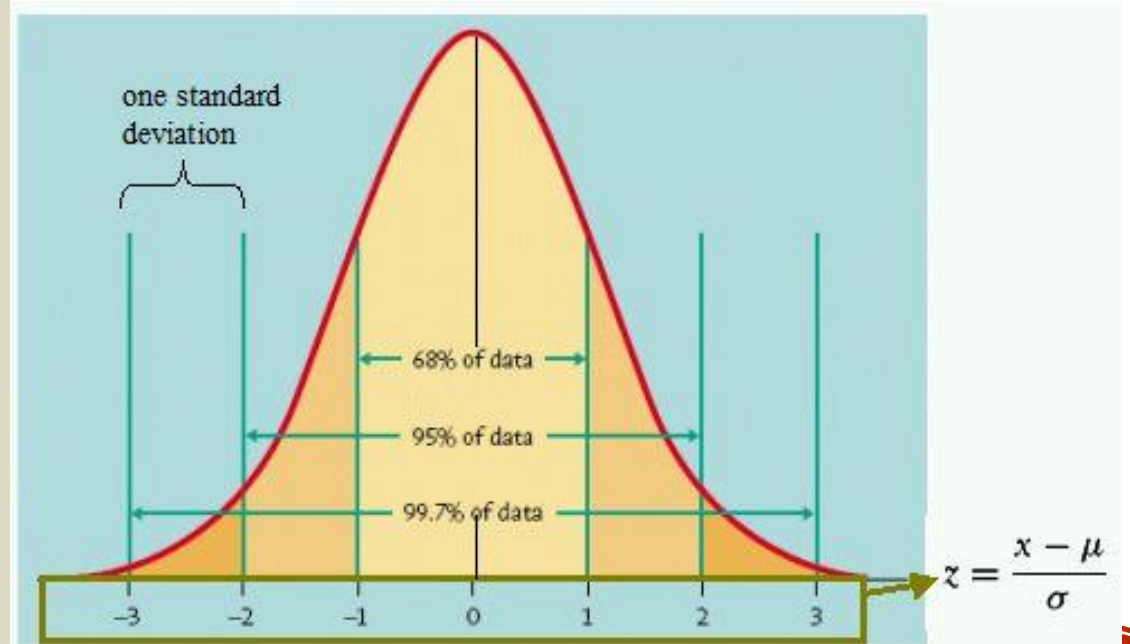
- It is a generalization of the binomial distribution where $n \rightarrow \infty$
- Instead of n and p , the parameters are the mean (μ) and standard deviation (σ).
- It really **is** bell-shaped since x is continuous and goes infinitely in each direction.
- The sum of independent normally distributed variables is normal.

INTERPRETING THE NORMAL DISTRIBUTION

Tight bounds on probability follow for Z-scores from normally distributed random variables:

IQ is normally distributed, with mean 100 and standard deviation 15.

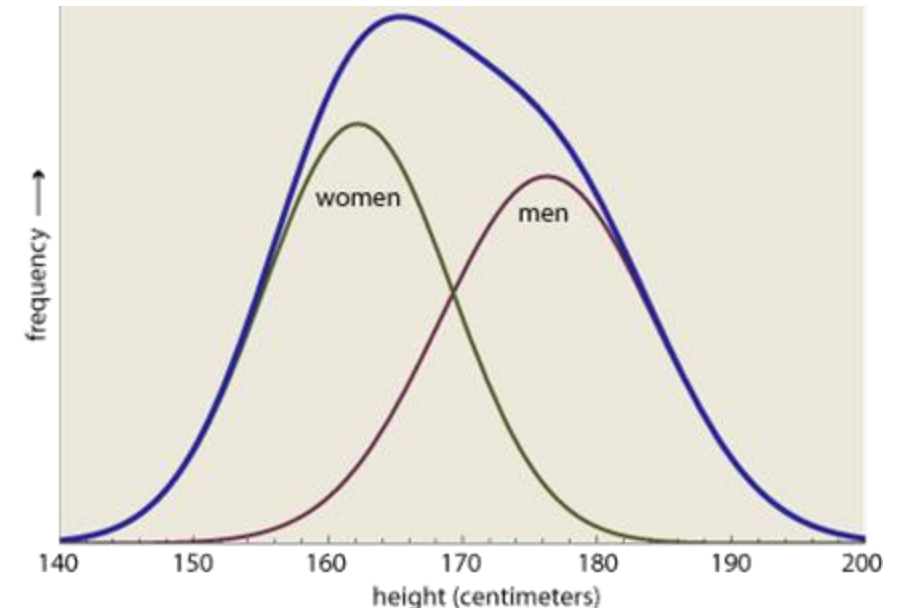
Thus about 2.5% of people have IQs above 130.



WHAT'S NOT NORMAL?

- Not all bell-shaped distributions are normal (i.e. stock returns are log normal with fat tails).
- **Mixtures of normal distributions are not normal, like full population heights.**

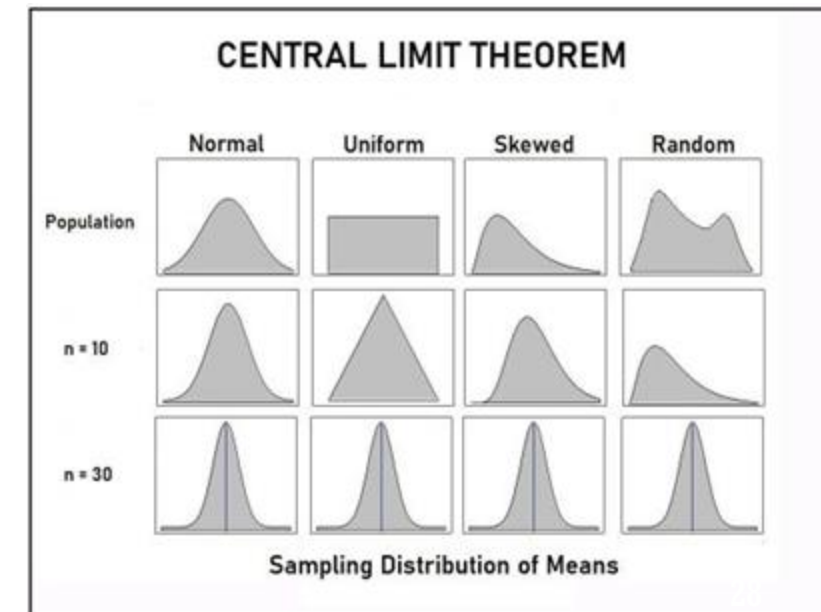
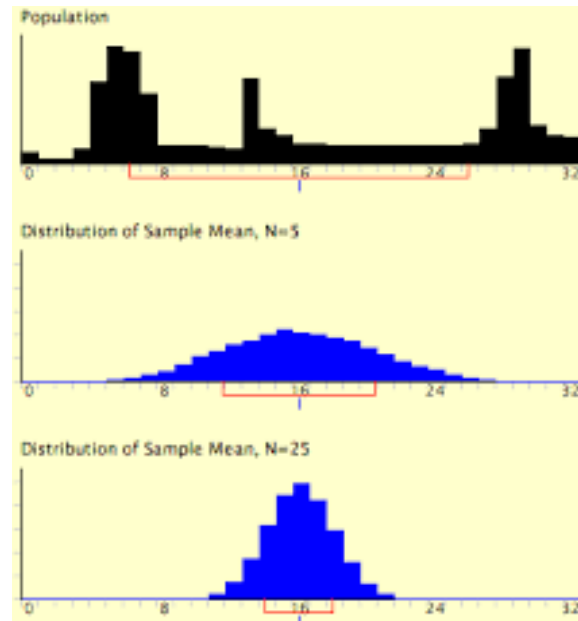
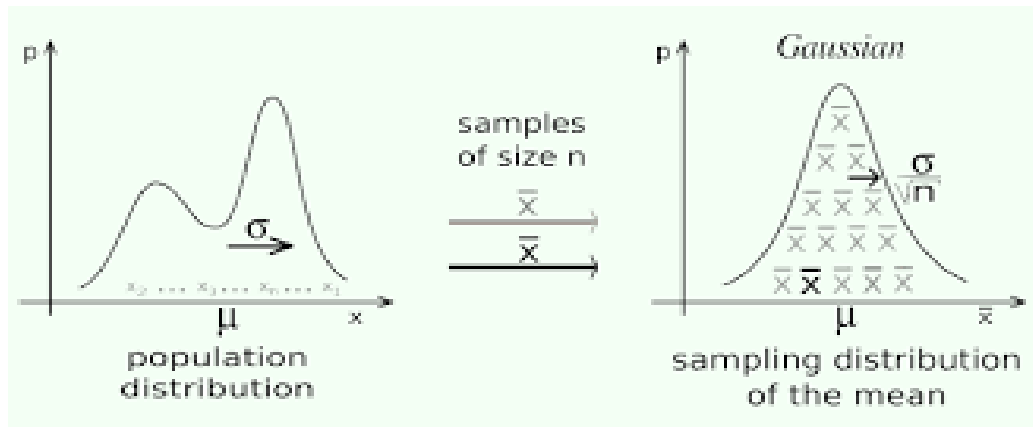
Statistical tests exist to establish whether data is drawn from a normal distribution, but populations are generally mixtures of multiple distributions: height, weight, IQ

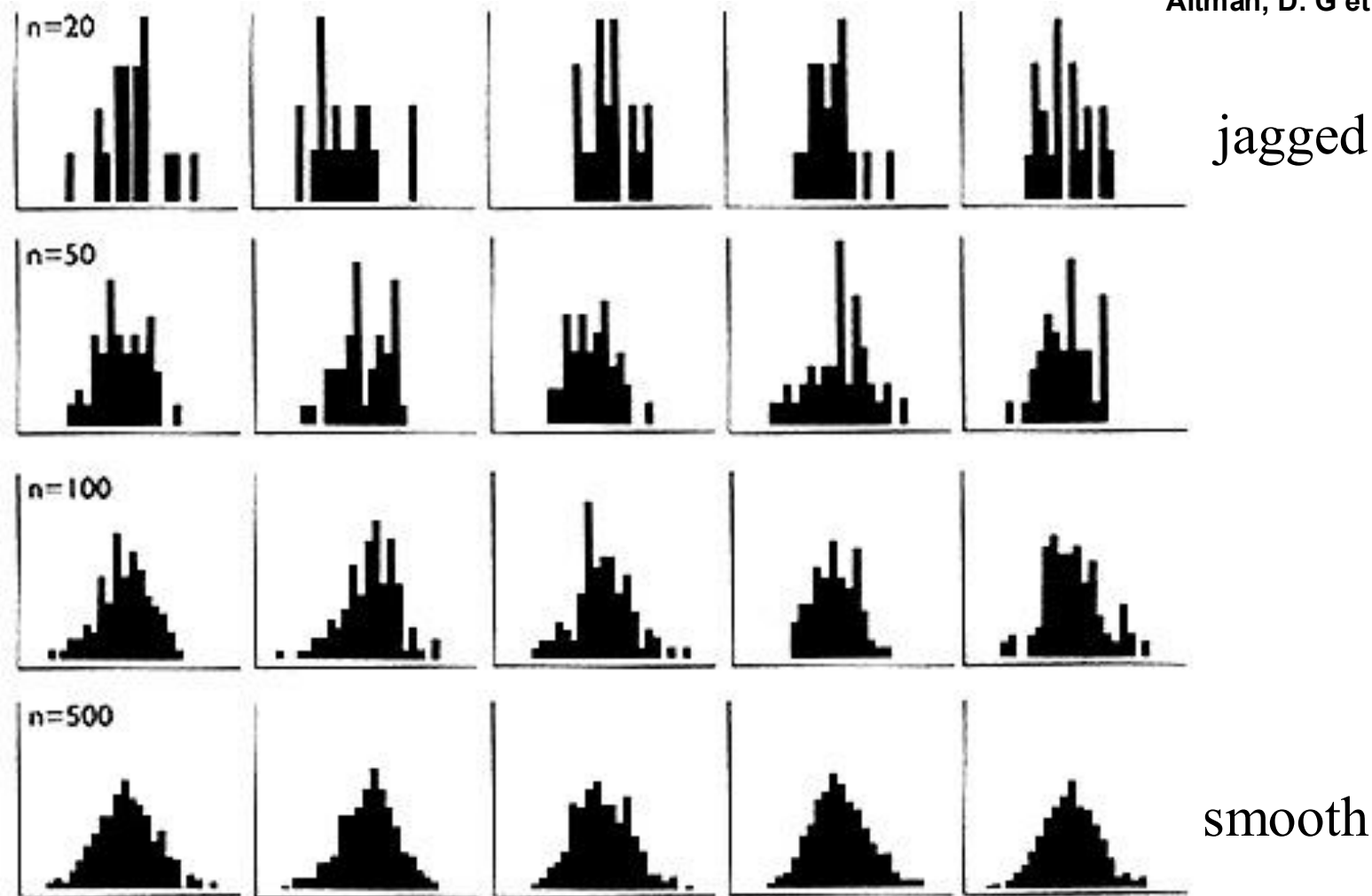


THE CENTRAL LIMIT THEOREM (CLT)

The mean of n **independent** samples from **any** distribution converges to the normal distribution.

If the population distribution has mean μ and SD σ , the mean distribution of n samples has mean μ and SD $\sigma/\sqrt{n}$



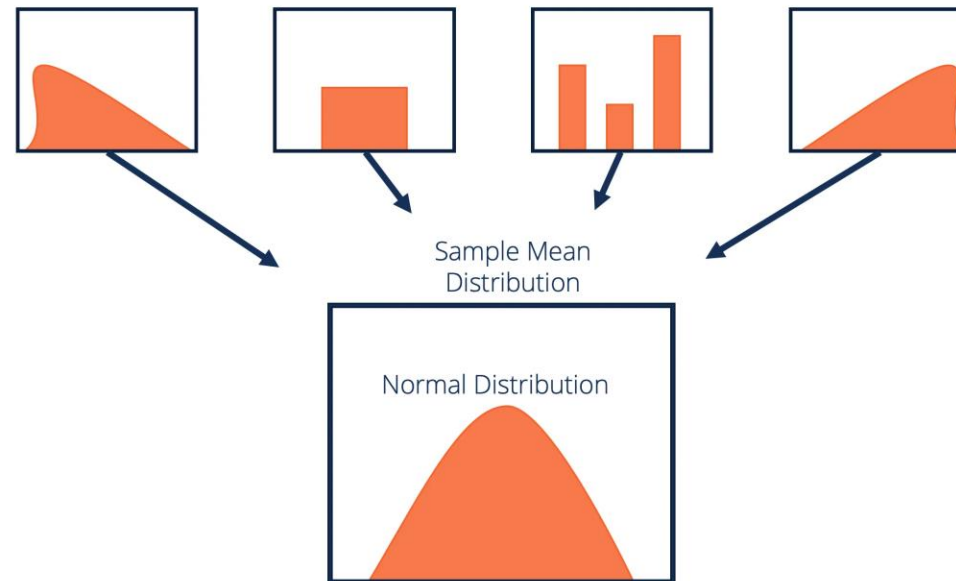


Central Limit Theorem: the larger the sample size, the closer a distribution will approximate the normal distribution or

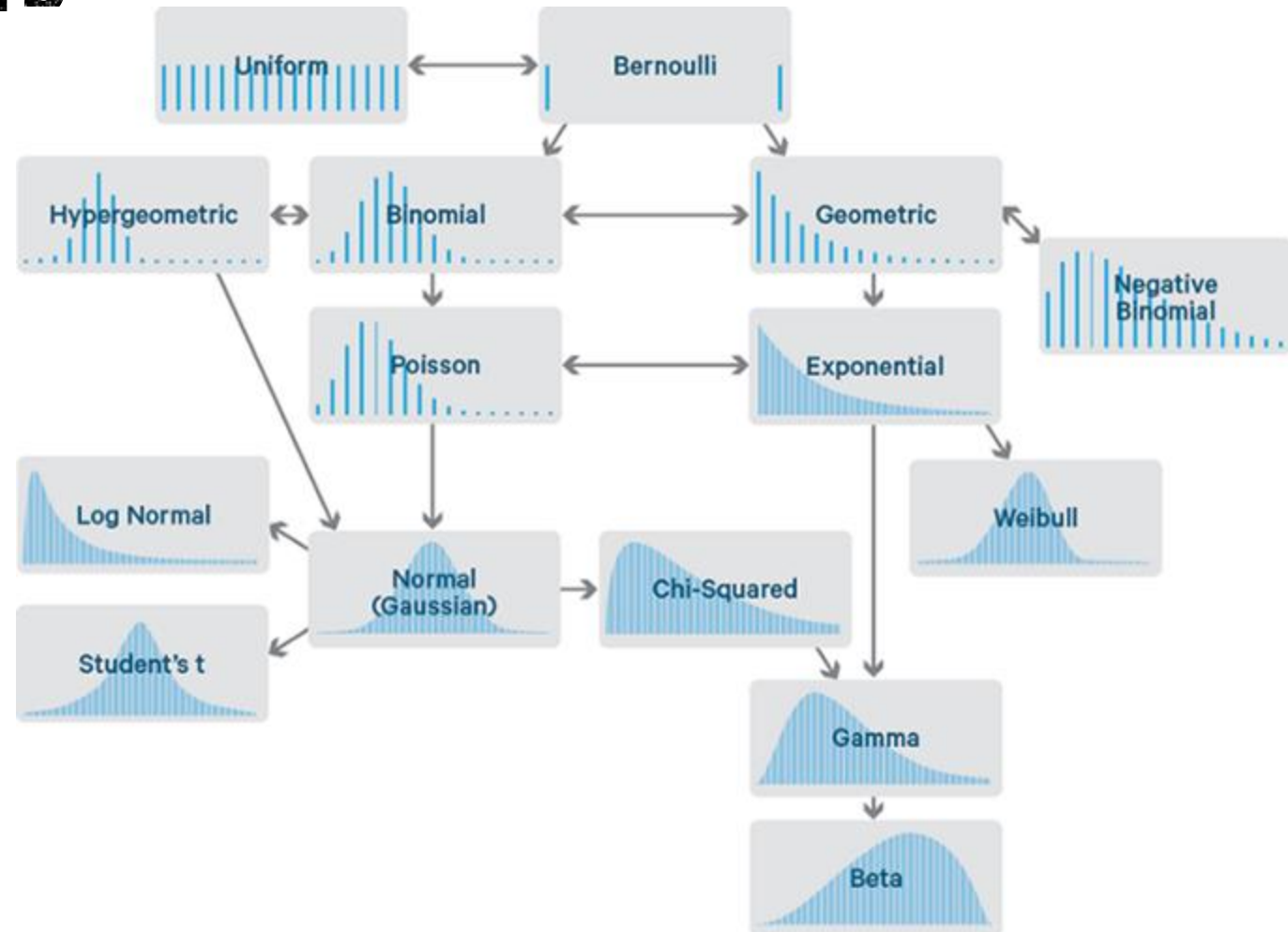
A distribution of scores taken at random from any distribution will tend to form a normal curve

THE CENTRAL LIMIT THEOREM

- This fundamental statistical principle allows for the application of normal distribution-based methods (like confidence intervals and hypothesis tests) to a wide range of problems, even when the underlying population is not normally distributed.

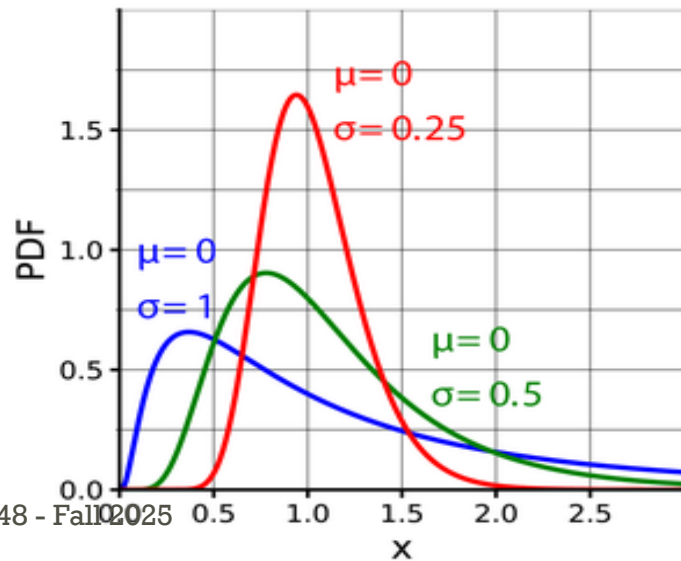


OTHER WEIRD PROBABILITY DISTRIBUTIONS

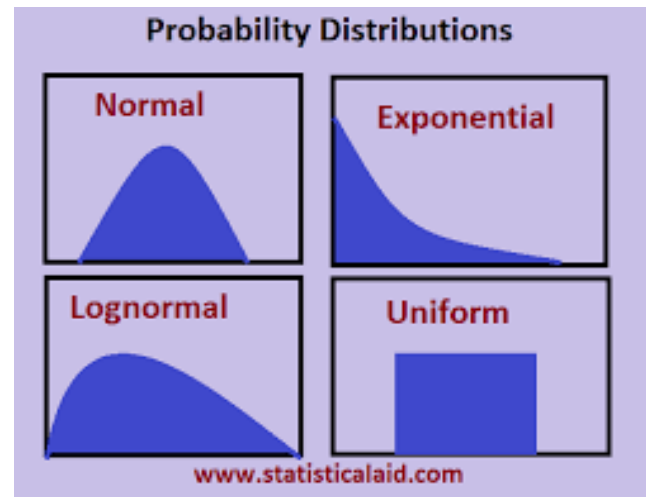


THE LOG NORMAL DISTRIBUTION

- This is a distribution for a random variable X such that the distribution of $Y=\text{Log}(X)$ is normal.
- A right-skewed continuous probability distribution
- Common applications in finance



CS4048 - Fall 2025



PROBABILITY DISTRIBUTIONS IN PYTHON

- Python library: `scipy.stats` (<https://openstax.org/r/scipy>)
 - `import scipy.stats`
- Probability density functions available within Python:
 - `binom()`—calculate probabilities associated with the binomial distribution
 - `poisson()`—calculate probabilities associated with the Poisson distribution
 - `expon()`—calculate probabilities associated with the exponential distribution
 - `norm()`—calculate probabilities associated with the normal distribution



DATA SAMPLING METHODS

SAMPLING

- Sampling: obtaining a small sample s to represent the whole data set N
- Allow an ML algorithm to run in complexity that is potentially sub-linear to the size of the data
- Key principle: Choose a **representative** subset of the data
 - Simple random sampling may have very poor performance in the presence of skew
 - Develop adaptive sampling methods, e.g., stratified sampling

EXAMPLE

Consider a research study that wants to randomly select a group of college students from a larger population to examine the effects of exercise on their mental health outcomes. Using student ID numbers generated by a computer program, 100 participants from the larger population were randomly selected to participate in the study to achieve the desired accuracy. This process ensured that every student in the university had an equal chance of being selected to participate. The participants were then randomly assigned to either the exercise group or the control group. This method of random sampling ensures that the sample is representative of the larger population, providing a more accurate representation of the relationship between exercise and mental health outcomes for college students.

MOTIVATION

- Data sampling serves four purposes.
 - To reduce the number of instances submitted to a DM/ML algorithm.
 - To support the selection of only those cases in which the response is relatively homogeneous.
 - To assist regarding the balance of data and occurrence of rare events.
 - To divide a data set into three data sets to carry out the subsequent analysis of DM/ML algorithms.

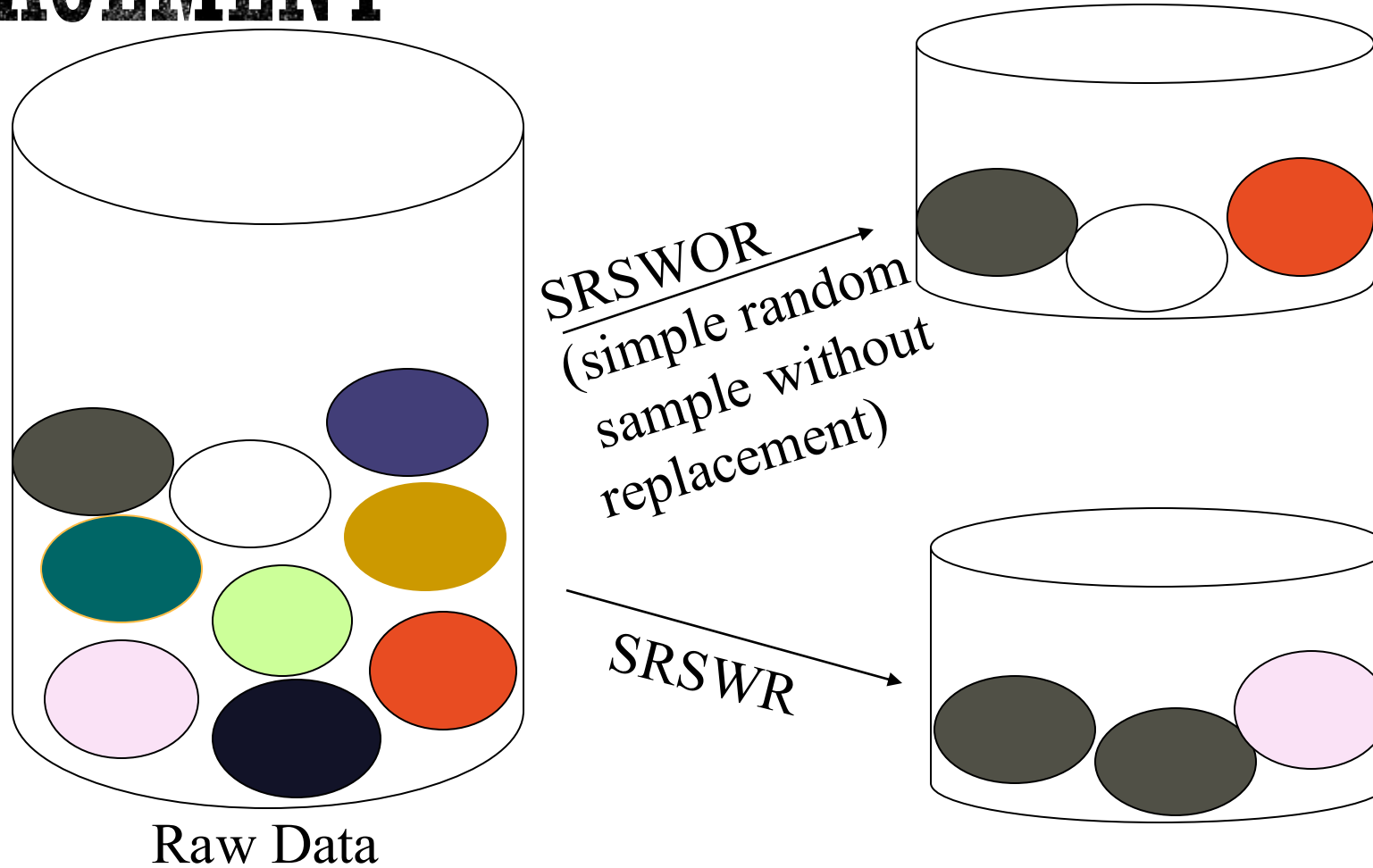
TYPES OF SAMPLING

- **Simple Random Sampling (SRS)**
 - There is an equal probability of selecting any particular item
- **Simple Random Sampling without replacement (SRSWOR)**
 - Once an object is selected, it is removed from the population
 - That is, drawing s of the N tuples from T ($s < N$), where the probability of drawing any tuple in T is $1/N$, that is, all examples have equal chance to be sampled.
- **Simple Random Sampling with Replacement (SRSWR)**
 - A selected object is not removed from the population
- **Cluster sampling:**
 - Partition the data set into mutually disjointed groups or clusters, then perform SRS

TYPES OF SAMPLING

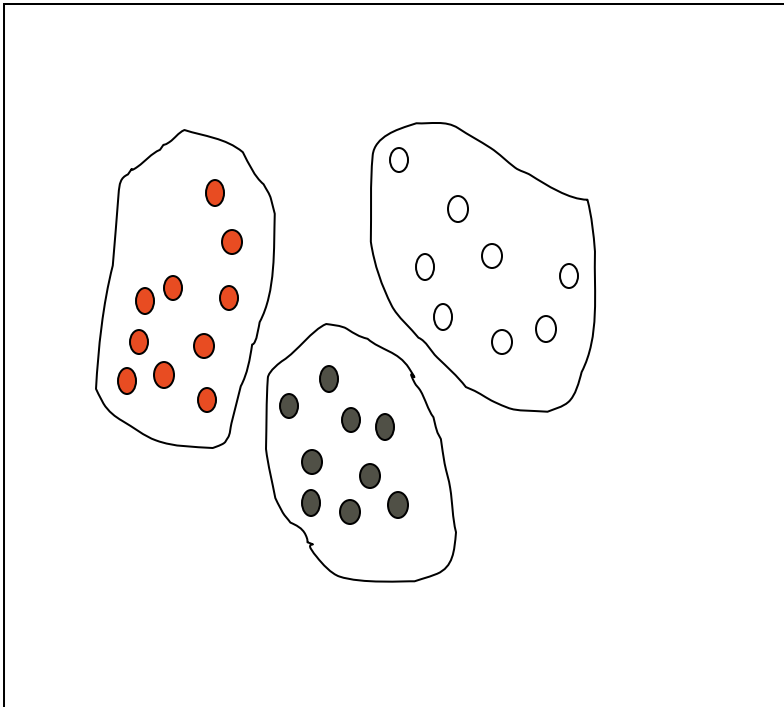
- **Stratified sampling:**
 - Partition the dataset into mutually exclusive sets (a.k.a. *strata*), and draw samples from each stratum (proportionally, i.e., approximately the same percentage of the data)
 - Used in conjunction with skewed data
 - Mostly used in classification tasks

SAMPLING: WITH OR WITHOUT REPLACEMENT

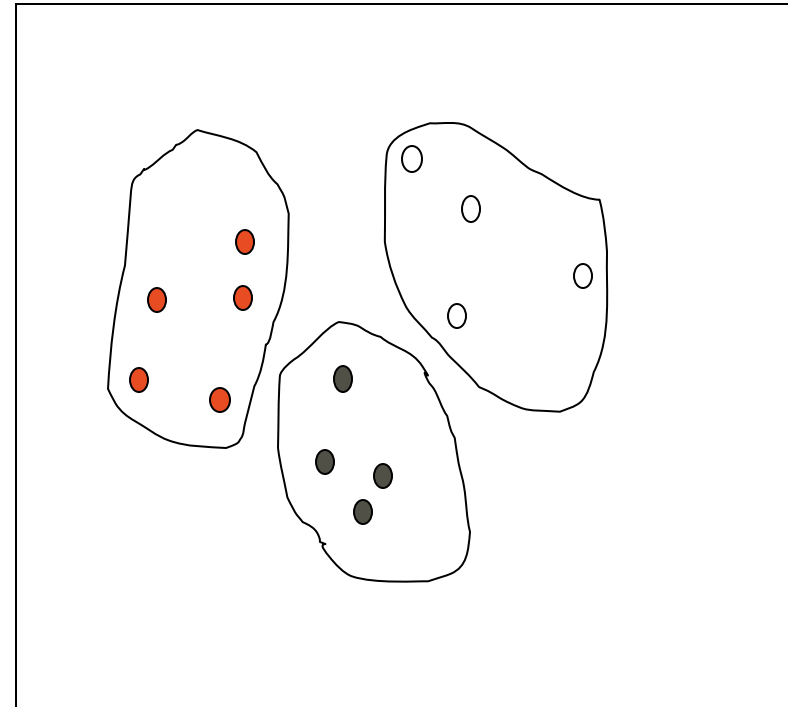


SAMPLING: CLUSTER OR STRATIFIED SAMPLING

Raw Data



Cluster/Stratified Sample



SAMPLING FOR SELECTING SPECIFIC CASES

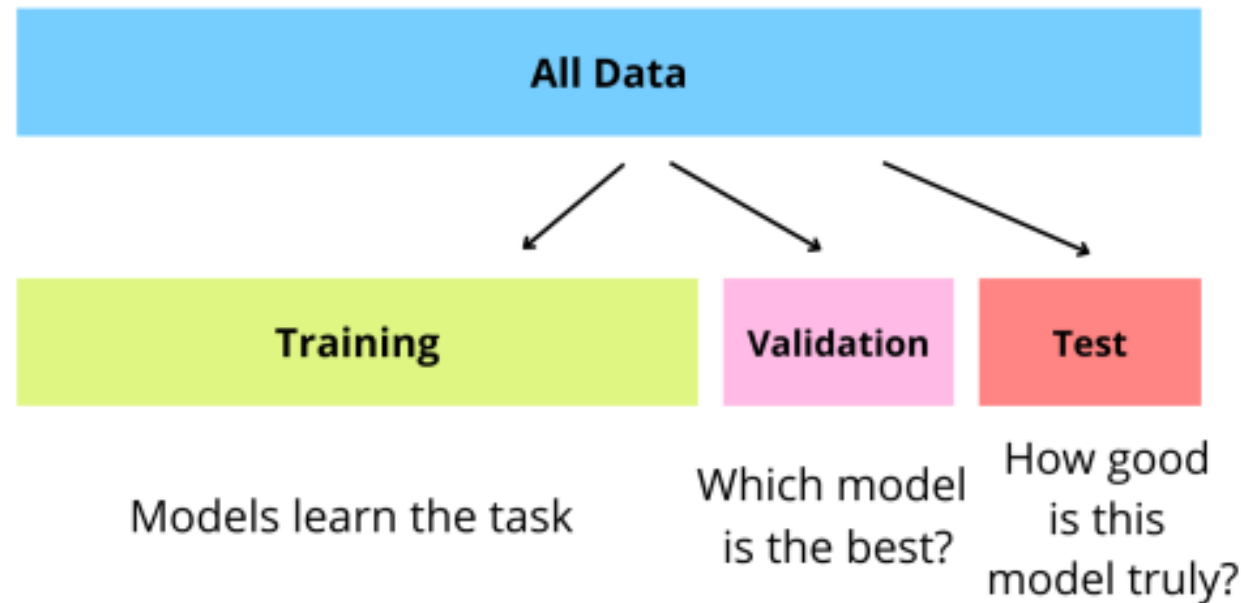
- Useful if some distinct patterns are clearly observable in a dataset
- Partition the data with different types of modeling
- Example:
 - Imagine the learning of the approving decision of bank loans depending on some economic characteristics of a set of customers. If data includes consumer loans and mortgages, it seems logical to partition both types of loans because the parameters and quantities involved in each one are completely different. Thus, it is a good idea to build separate models on each partition.

SAMPLING FOR DATA BALANCING

- Some predictive modeling methods are sensitive to imbalanced data (e.g., ANNs, decision tree)
- Imbalanced data: the target class is less representative for one category
 - Usually, the rare class is more important
 - Known methods to deal imbalanced datasets include **over-sampling** and **under-sampling**

SAMPLING FOR DATASET SPLITS

- A general approach is to split the data into **training and testing** datasets
- Another split can be performed within the training dataset to avoid model over-fitting
 - This data split is known as **validation** set



READING MATERIAL

- Chapter#3: Principle of Data Science
- Chapter#1, 2: Practical Statistics [Bruce]
- An article: <https://www.kdnuggets.com/2021/03/statistics-data-scientists-should-know.html>